DICOM Spatial Metrics
User Manual

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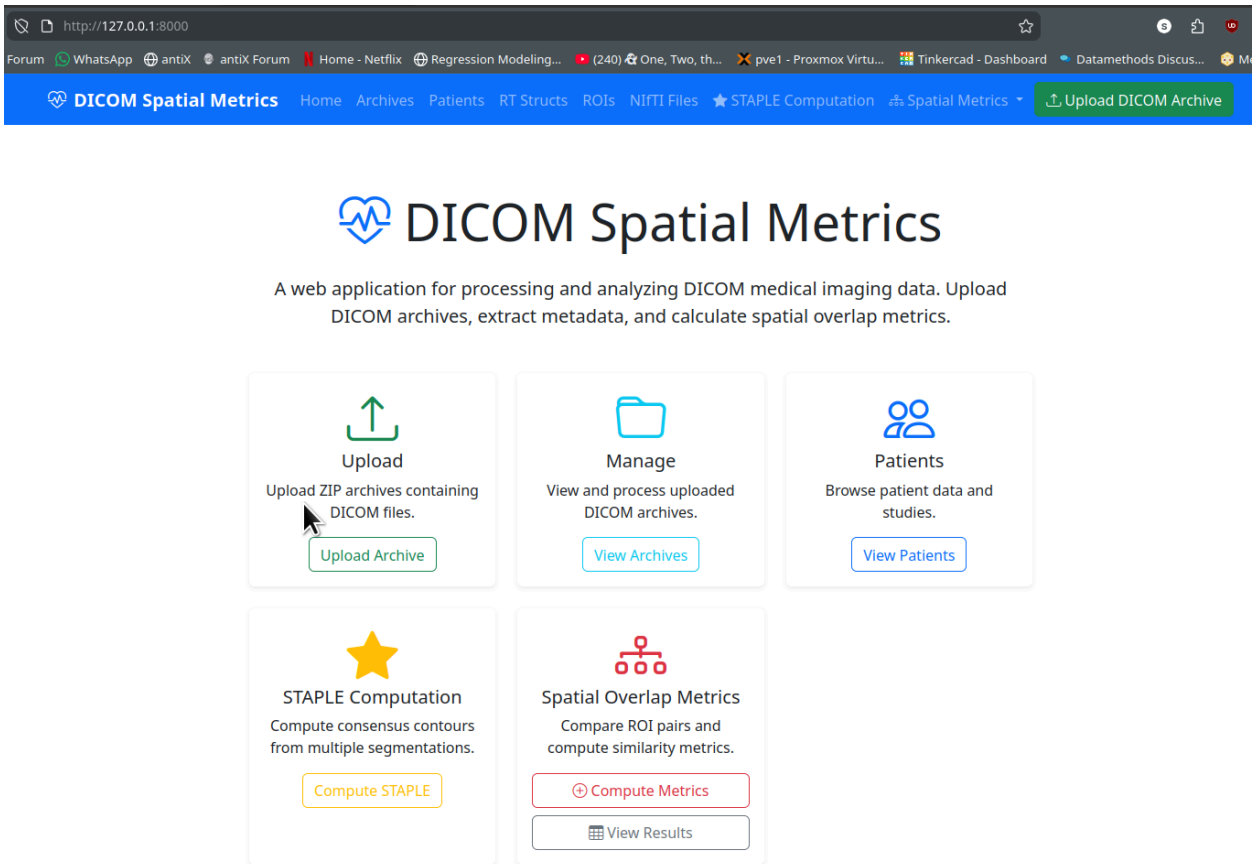
DICOM Spatial Metrics is an open source web application to allow users to compute a variety of spatial overlap metrics between different structures. It also allows computation of "consensus" contours using STAPLE algorithm. Please follow the instructions to use the software after installation

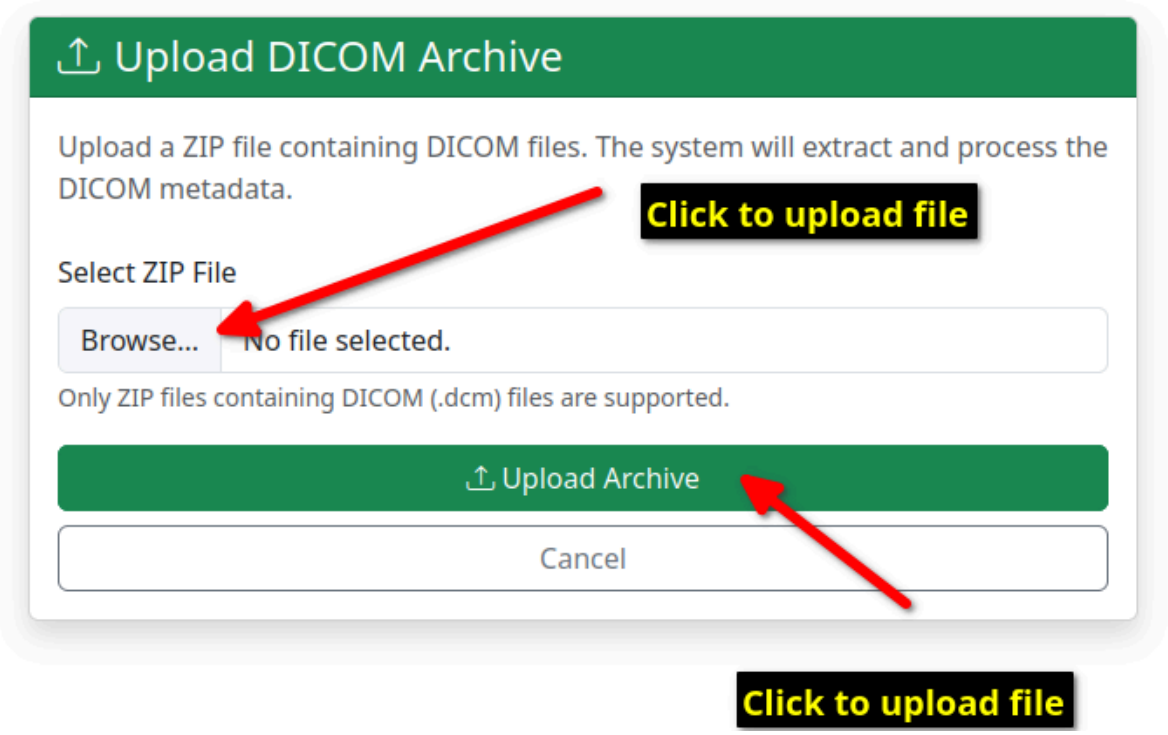
Upload a DICOM Archive

The first step is to upload a Zip file comprising of DICOM files. In order to do comparison, you need to have two or more structures at the minimum. However to compute STAPLE contours, the same structure should be segmented in multiple structuresets. A structureset is a special type of DICOM file with information on the segmentation performed. Note that the archive requires a zip file.

Open your browser and go to the url. For most cases, the URL will be like https://localhost:8001. Note that the 8001 is the port on which the webapp is running.



Click on Upload Archive button on the dashboard or the Upload DICOM archive button at the top right to begin.

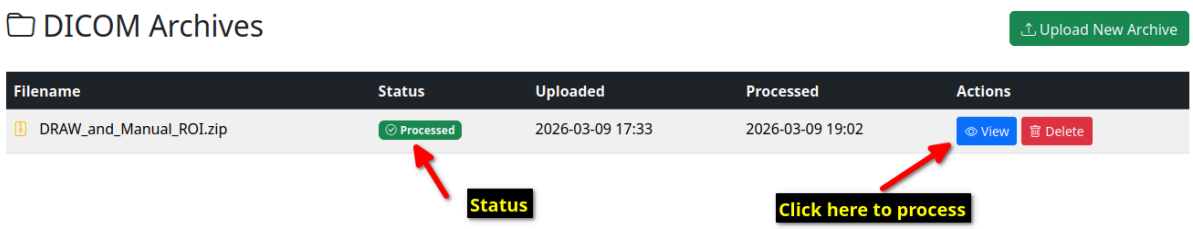


Click on the Browse button to select the zip file and click upload archive to upload the archive (zip) file containing the dicom files. Note that you can upload a single zip file but each zip file can contain multiple patients' data.

Note that the dicom files will be uploaded on a folder called dicom_zip_files under the media directory.

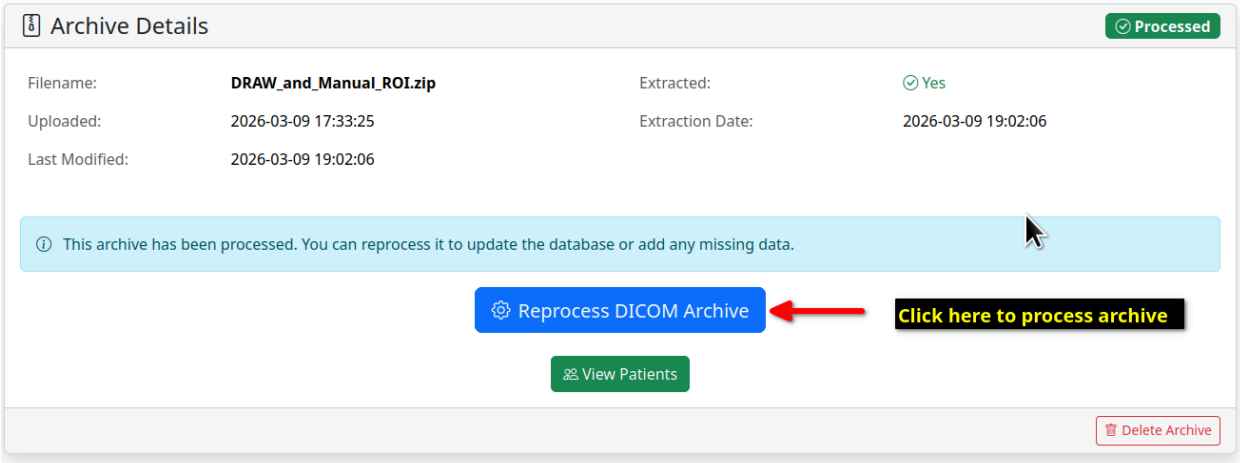
Process DICOM Archive

The second step is to process the DICOM archive to populate the data in the DICOM file. This step will extract all files from the zip archive, validate the DICOM files and save them in a proper DICOM format on the hard drive. The files will be stored in a folder called dicom_files in the project root.



When the DICOM archive file has been uploaded for the first time, it will show as unprocessed and you can click the View button to go to the page with the details of the archive to process the file. If the file has been processed once it will show processeed in the table but you can reprocess it again.

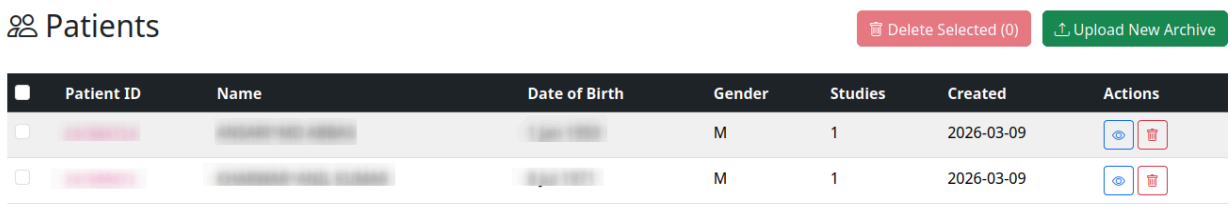
Clicking on the Delete button will delete the database entry not the zip file !



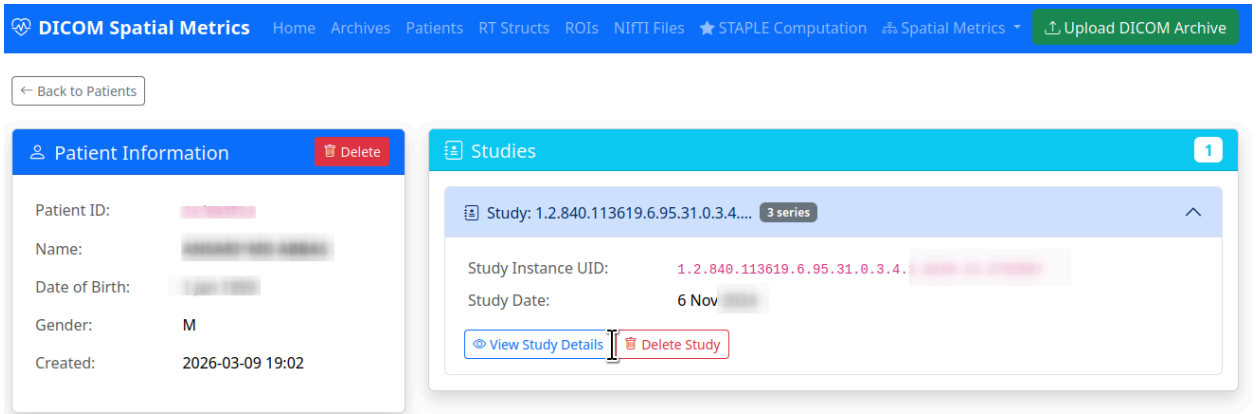
In the screenshot above, as the archive has been processed, the buton says reprocess DICOM archive. However if this has never been processed, it will show Process DICOM Archive instead. Clicking this button will process the archive by extracting the files, reading metadata one by one, and finally saving it as a valid DICOM file. Note that files which are not DICOM files will not be saved.

Patient Page

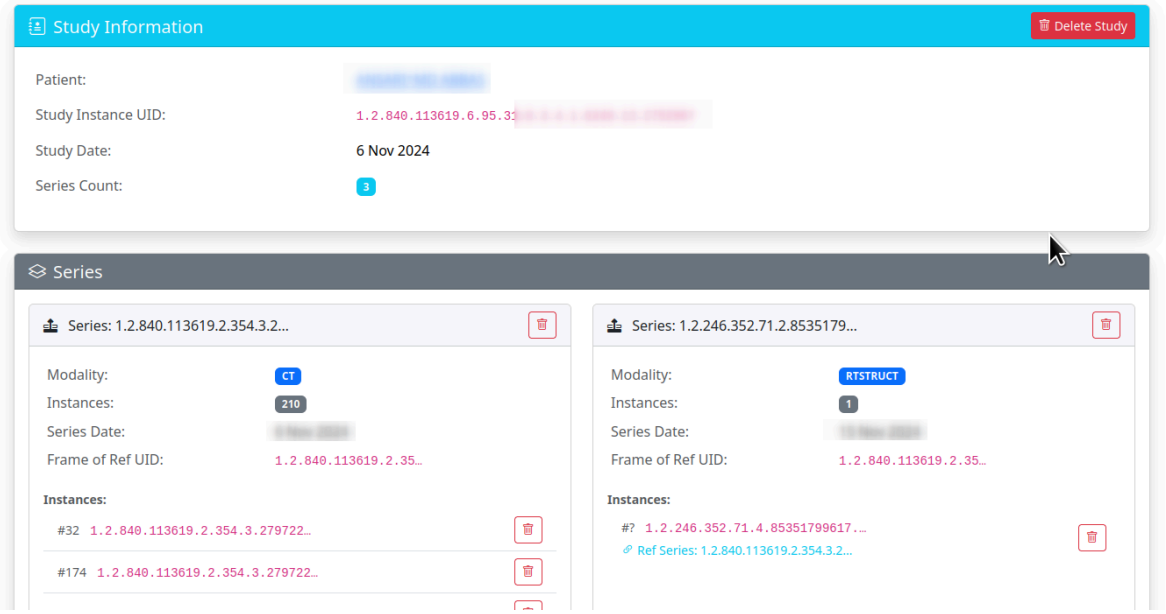
The page for the patients can be accessed by clicking the link on the top navigation bar which says Patients. Clicking this opens a tabular view of the patients in the system. Note that this does not show the patients belonging to a specific ZIP file archive uploaded but for all patients in the database.



Clicking on the blue eye icon will allow you to visualize the details for the given patient. The checkbox is to allow you to delete the patient information from the table. This data is automatically processed from the DICOM file uploaded.



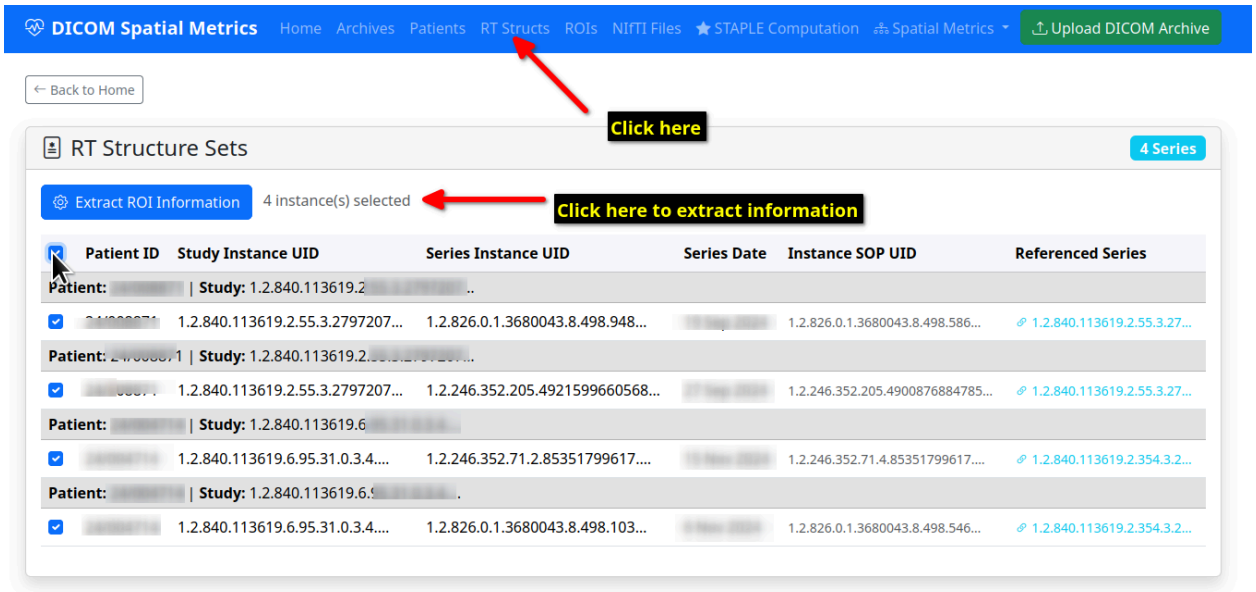
Clicking on the blue View Study Details button will show a page which has details of the patient's studies. Each study can have multiple series associated with it. For example, if the patient has undergone a full bladder and empty bladder scan and contours have been done on the full bladder scan then the patient will have two studies - Full Bladder and Empty Bladder. In the Full bladder study you will find two series - one will be the CT scan and the other will be the structureset. IN the Empty bladder study you will have a single series.



Caution:
Deleting a patient deletes the associated Studies, Series and Instances. Deleting a Study deletes the associated series and Instances. Deleting a Series will delete the associated Instances.
These data are linked as follows:
Patient > Study > Series > Instance
Note that the files are also deleted from the disk.

Extract ROI Information

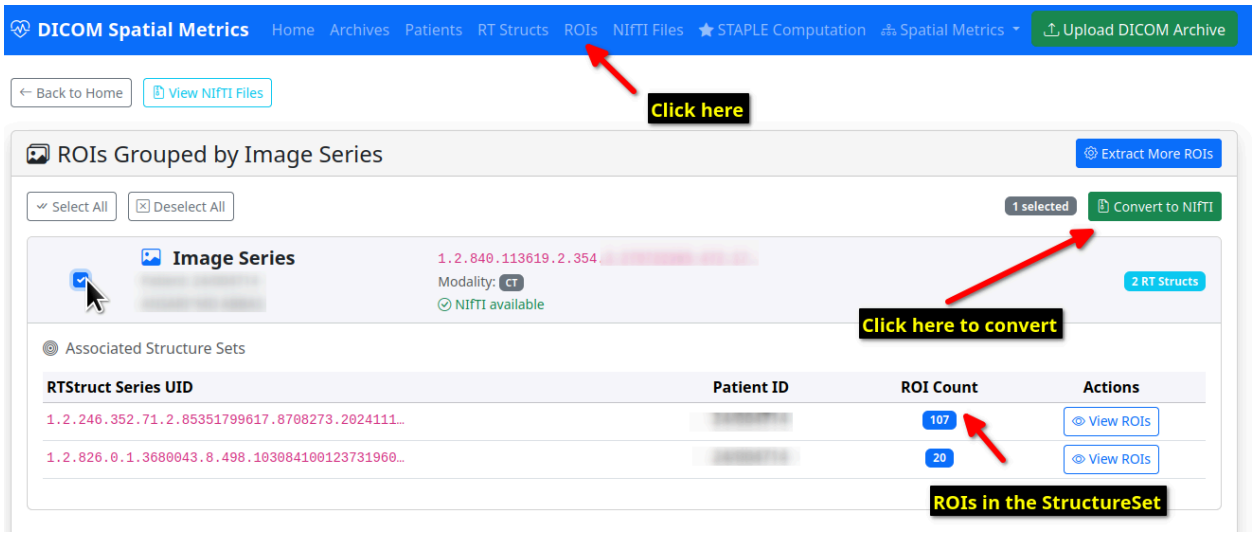
After the DICOM archive has been processed you can extract the ROI information contained in the structureset files. This is necessary for the next step of conversion of the DICOM files to NIFTI files.



Clicking the Extract ROI information will extract all available ROI information (name and the structure number) from the DICOM files and make it available for processing in the NIFTI files. You can click the ROIs link to perform the next step before computing the spatial metrics.

Processing ROIs to NIFTI

In order to proceed with computation of the spatial overlap metrics and the STAPLE contour, we need to convert the ROIs to NIFTI files. To do this please go to the ROIs page which shows the list of structure sets available for a given patient.

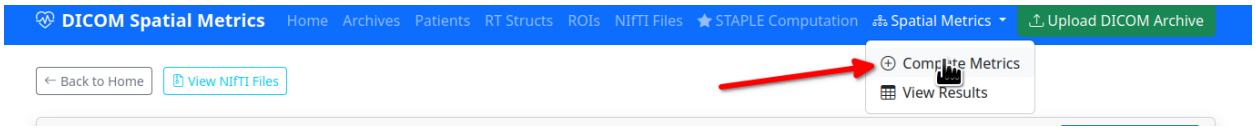


You can select multiple patients at once to convert to NIFTI format. The task will show a progress bar with the progress of conversion. The CT scan will be converted into a single nifti file and each ROI will have a separate nifti files. All nifti files will be compressed to reduce disk space utilization and faster network transfer.

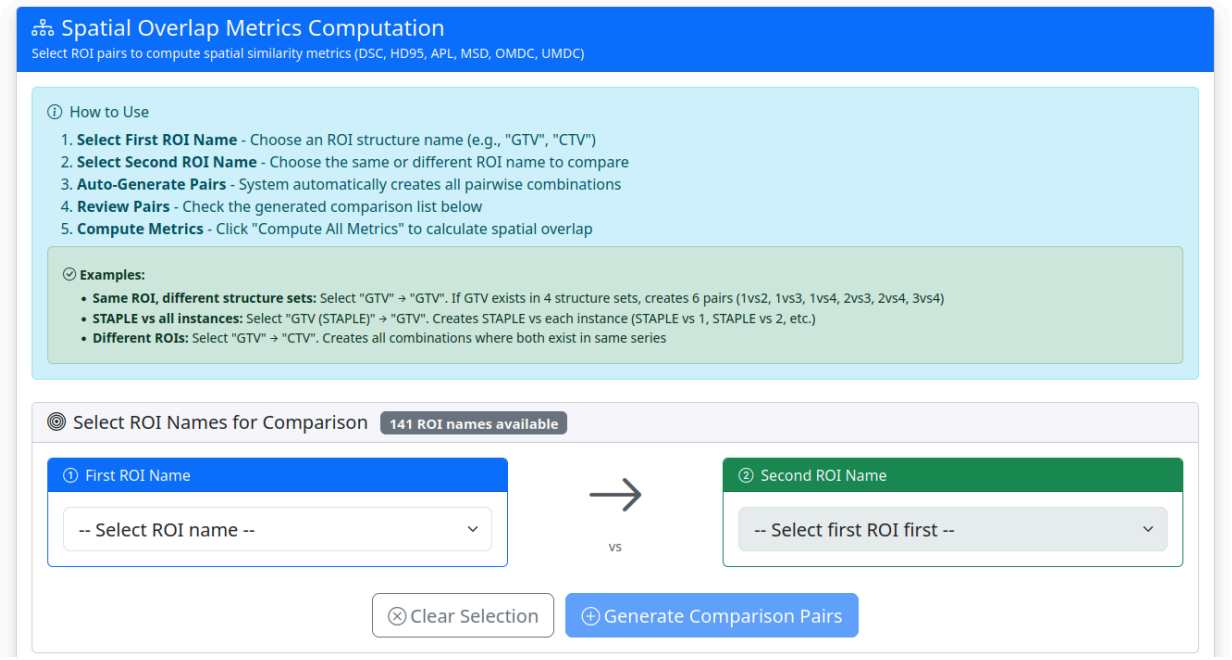
Nifti files are stored in the disc as a subdirectory under the media folder in the project root. Each patients nifti file are stored as Patient > Study > Series > Nifti

Compute Spatial Overlap

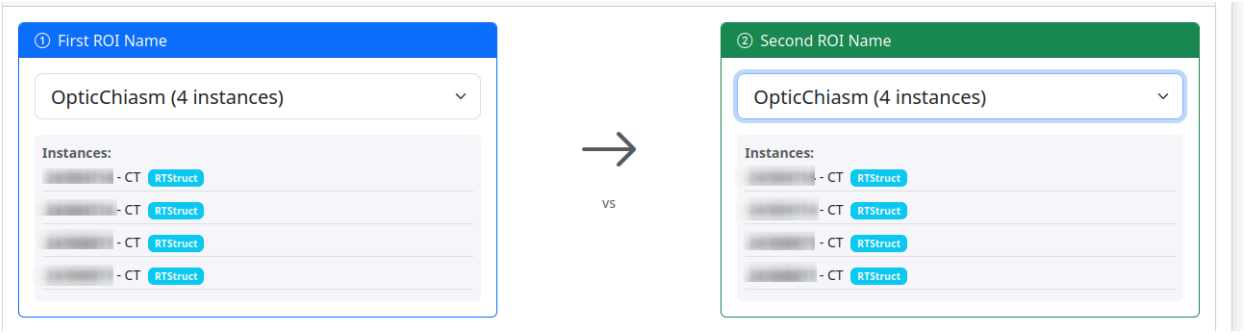
If you wish to proceed with spatial overlap calculation please click the Spatial metrics > Compute Metrics link on the navigation bar on the top.



This will open a page where you can visualize the ROIs which you can use for computation of spatial overlap metrics.

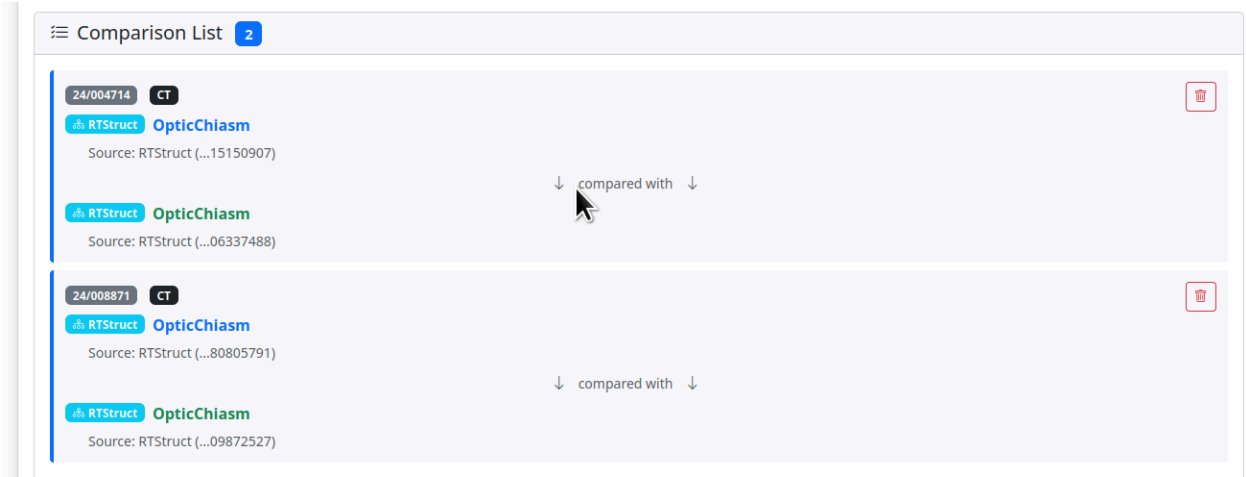


As the guidance on the page suggests, select the first ROI (reference ROI) and the second ROI which is to be compared against the reference ROI.

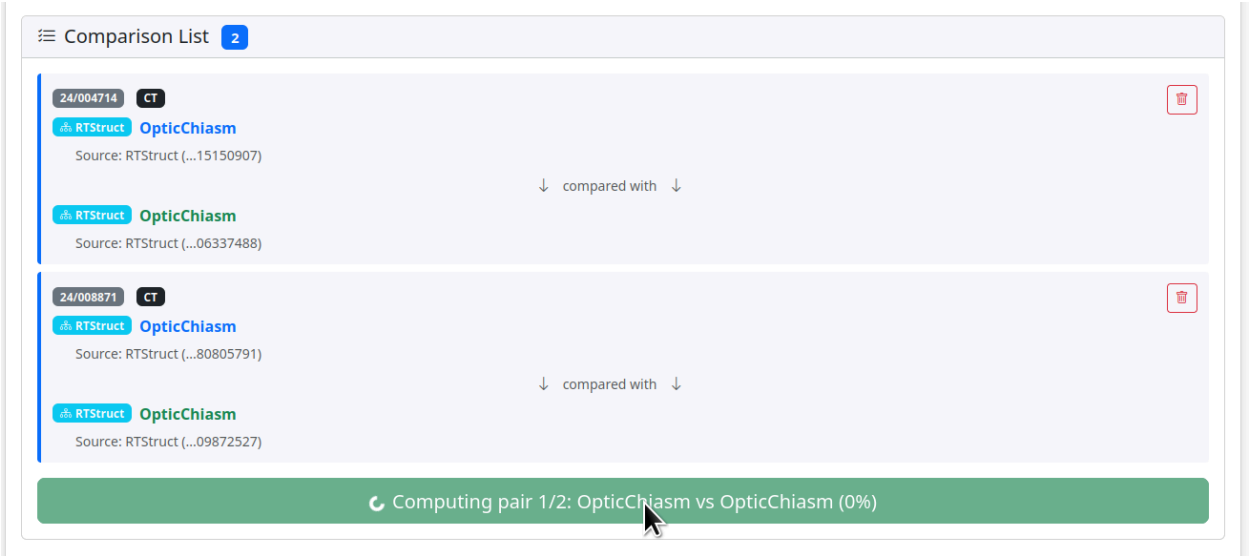


In the above screenshot, optic chiasm has 4 instances available, as there are 2 patients each of whom have 2 contours of optic chiasm.

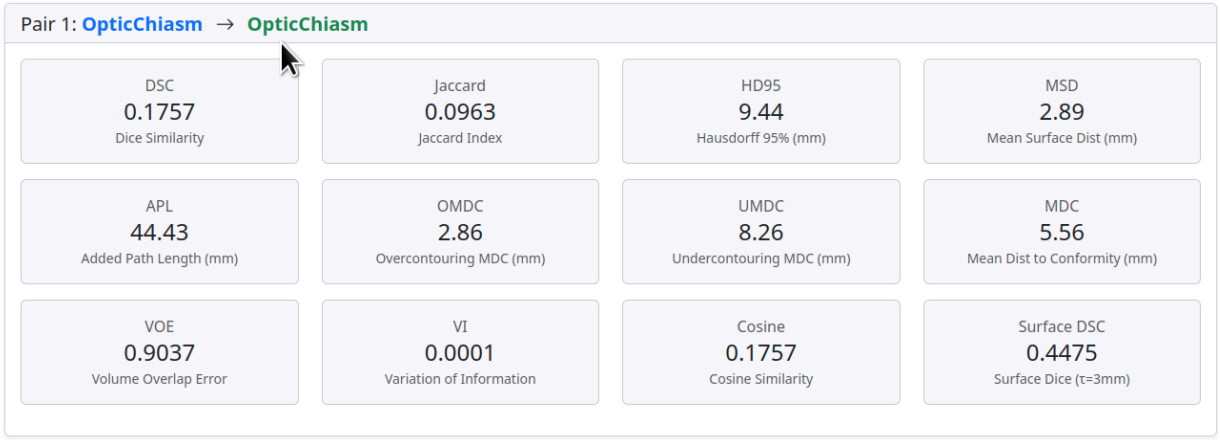
Click the button to generate comparison pairs. Note the comparison pairs are generated for an individual series. In this case there will be two comparison pairs as there are 2 series each with 2 contours.



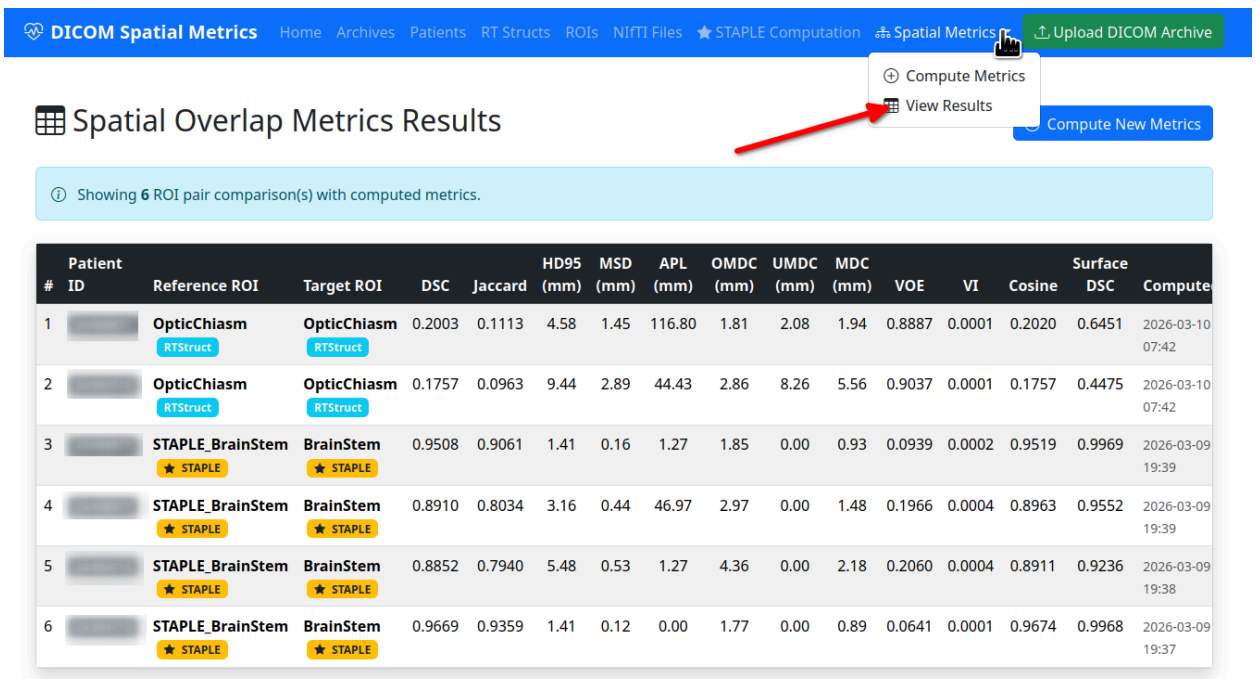
Note that in this case there are no STAPLE contours. If STAPLE contours were available then there would be 4 possible comparisons.



Click on the green button below the pairs to compute the metrics. After the computation is completed you can see the results in a grid for each pair of comparison. The computed metrics are: Dice Similiarity Coefficient, Jaccard Coefficient, 95% percentile of Haurdorff distance, MSD, Added path length, Overcontouring mean distance to conformity, Undercontouring mean distance to conformity, Mean distance to conformity, Volume overlap error, variation of information, Cosine similarity and Surface DSC.



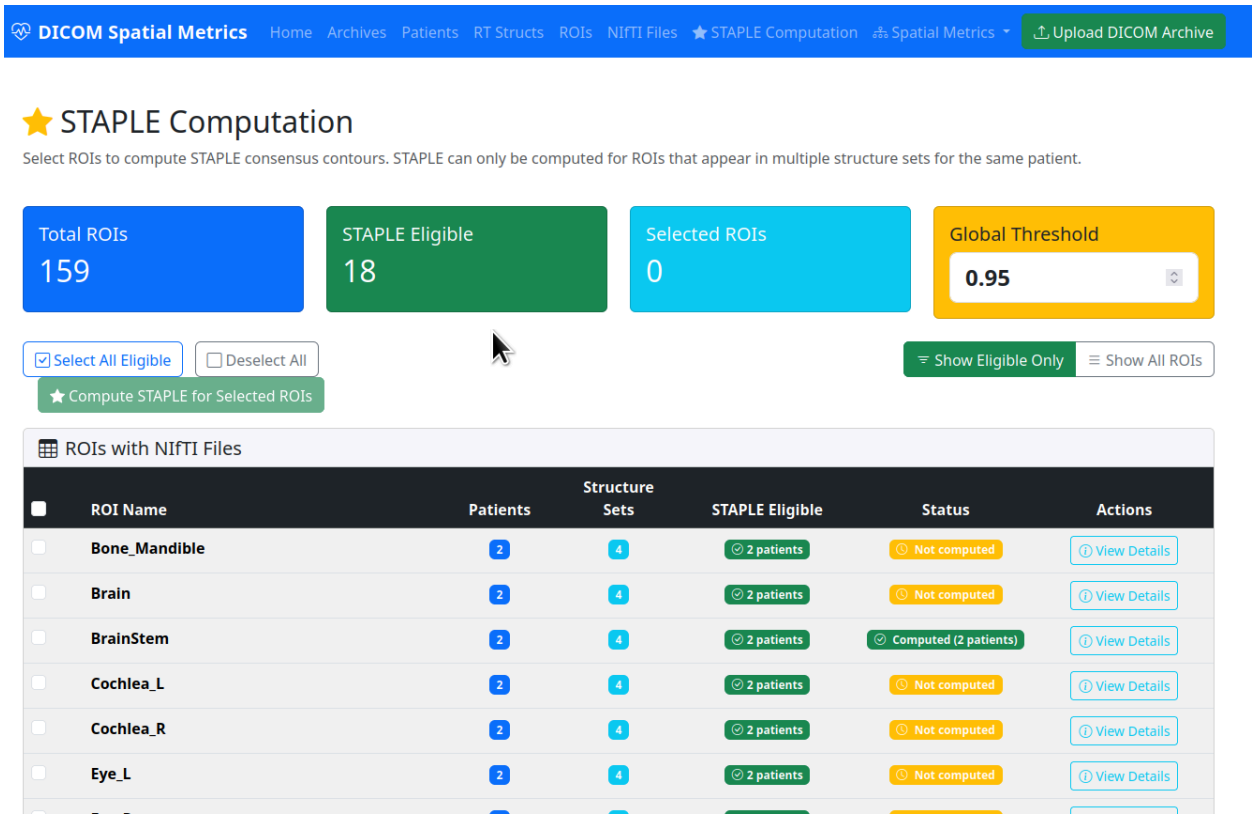
Note that the results will also be available in dedicated page for the spatial overlap metrics results.



#	Patient ID	Reference ROI	Target ROI	DSC	Jaccard	HD95 (mm)	MSD (mm)	APL (mm)	OMDC (mm)	UMDC (mm)	MDC (mm)	VOE	VI	Cosine	Surface DSC	Compute
1	[redacted]	OpticChiasm RTStruct	OpticChiasm RTStruct	0.2003	0.1113	4.58	1.45	116.80	1.81	2.08	1.94	0.8887	0.0001	0.2020	0.6451	2026-03-10 07:42
2	[redacted]	OpticChiasm RTStruct	OpticChiasm RTStruct	0.1757	0.0963	9.44	2.89	44.43	2.86	8.26	5.56	0.9037	0.0001	0.1757	0.4475	2026-03-10 07:42
3	[redacted]	STAPLE_BrainStem STAPLE	BrainStem STAPLE	0.9508	0.9061	1.41	0.16	1.27	1.85	0.00	0.93	0.0939	0.0002	0.9519	0.9969	2026-03-09 19:39
4	[redacted]	STAPLE_BrainStem STAPLE	BrainStem STAPLE	0.8910	0.8034	3.16	0.44	46.97	2.97	0.00	1.48	0.1966	0.0004	0.8963	0.9552	2026-03-09 19:39
5	[redacted]	STAPLE_BrainStem STAPLE	BrainStem STAPLE	0.8852	0.7940	5.48	0.53	1.27	4.36	0.00	2.18	0.2060	0.0004	0.8911	0.9236	2026-03-09 19:38
6	[redacted]	STAPLE_BrainStem STAPLE	BrainStem STAPLE	0.9669	0.9359	1.41	0.12	0.00	1.77	0.00	0.89	0.0641	0.0001	0.9674	0.9968	2026-03-09 19:37

STAPLE Contour Computation

In studies with multiple raters STAPLE contours provide a better way to determine the consensus as compared to naive majority voting. STAPLE contours can be computed only for ROIs which are same and present in multiple structureset files. In order to compute STAPLE contours click on the link to compute STAPLE contours (indicated with a star).



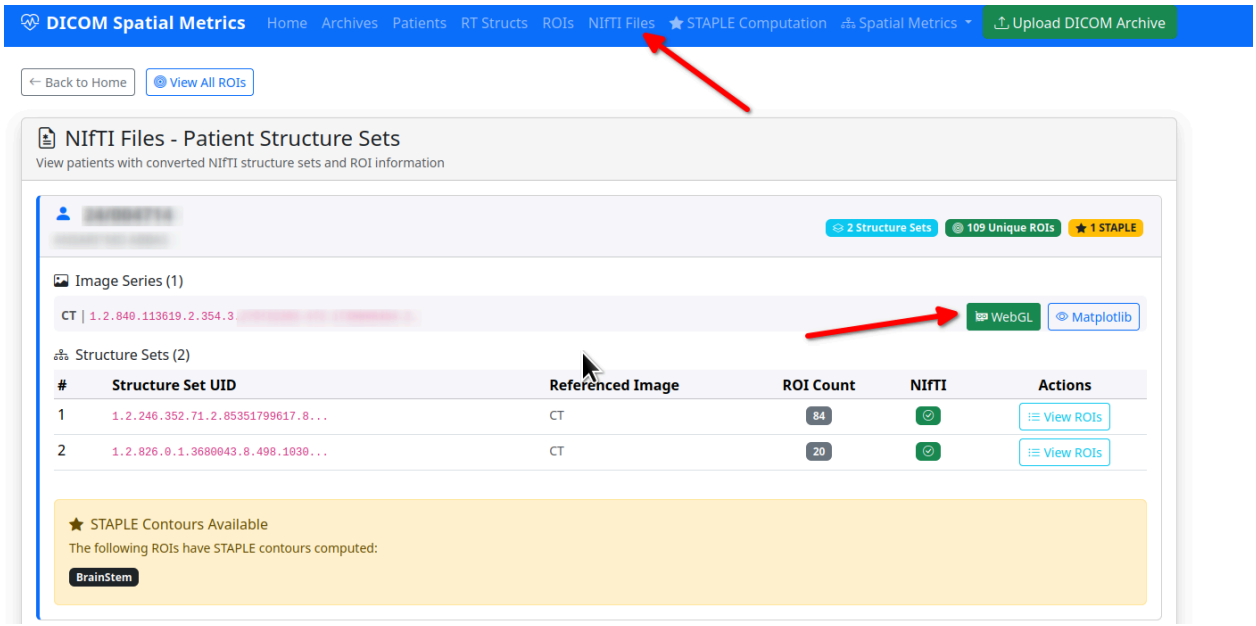
Note STAPLE eligible ROIs will be ROIs which have been extracted to NIFTI format and for ROIs where more than one contour are available. The global threshold button computes the STAPLE contour based on the agreement - in this case 95%. Click the green compute STAPLE for selected ROIs to compute STAPLE for multiple structures at once.

Once computed the table will display if the STAPLE contour is available for the specific ROI or not.

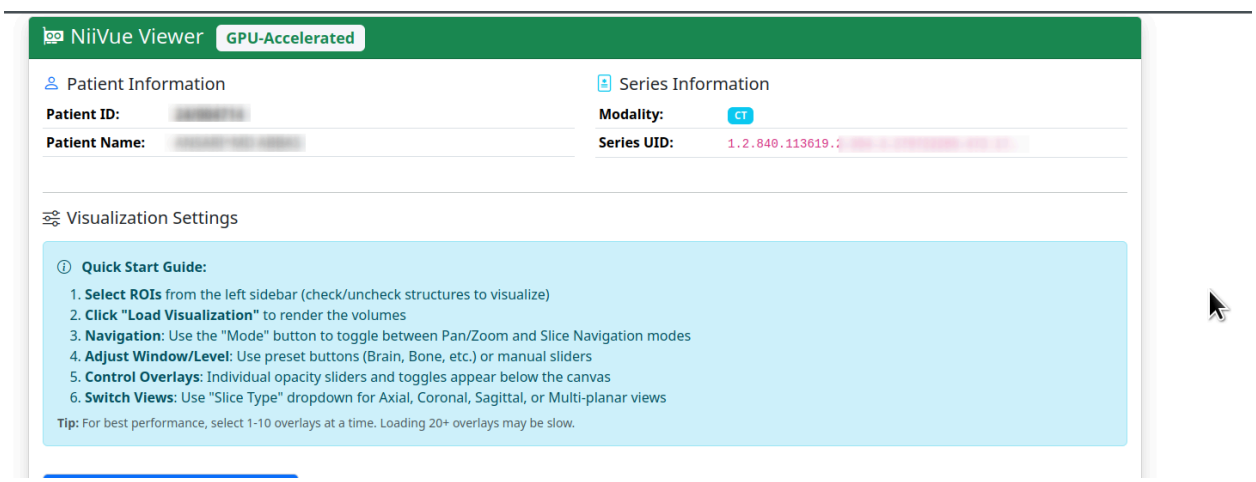
Click the view details to see which patients are contributing to the STAPLE contour.

Visualization of the NIFTI

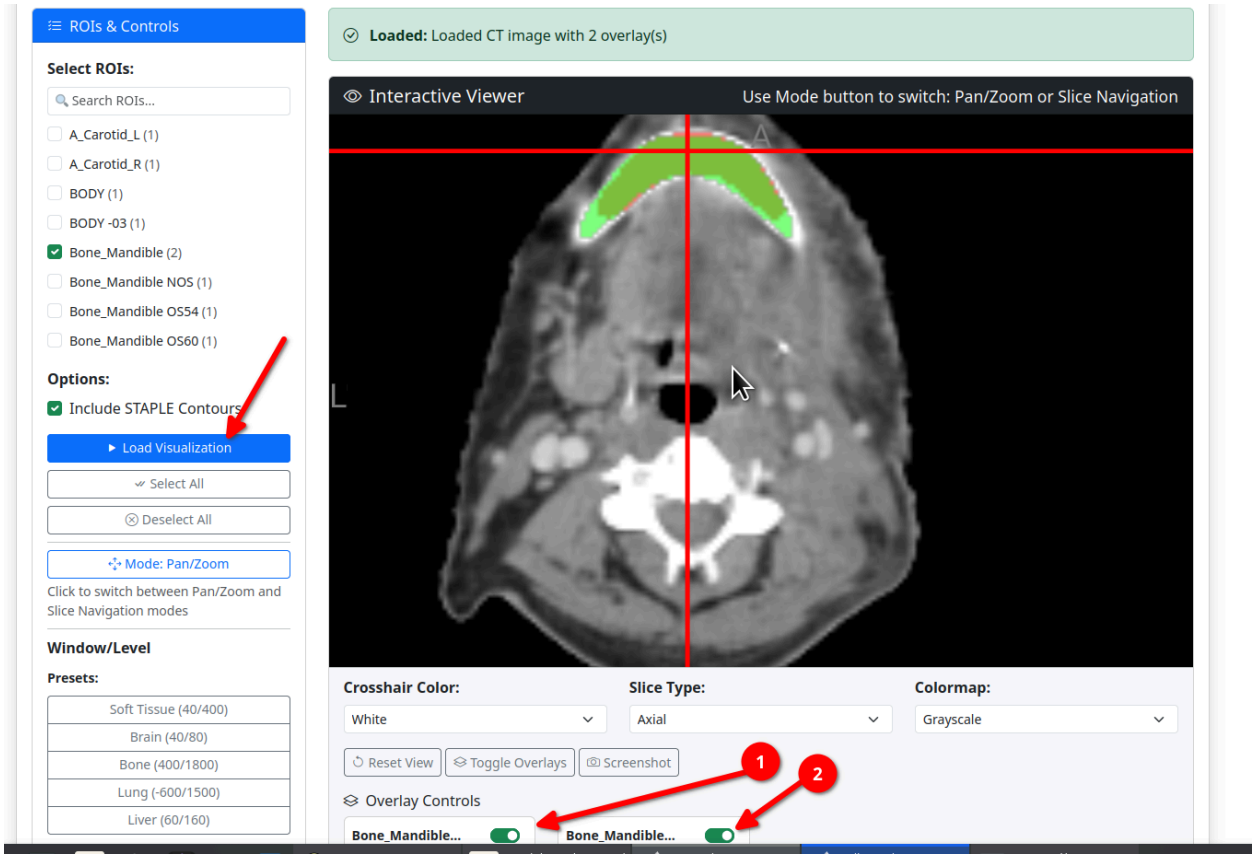
A visualization system is available if you wish to visualize the STAPLE contours.



This page open when the Nifti files link is clicked. Beside each image series you can see two buttons. For a modern computer with the latest Chrome / Firefox / Edge based browser we recommend using the WebGL viewer which should be supported for most modern computers. However if note then the Matplotlib viewer can be used (significantly slower).



Note that the viewer uses niiVue package in the backend.



Select a few structures and then click the Load visualization button to display the images. You can see the images in three planes. Use the Mode:Pan/Zoom button to change the mode to Zoom / Pan (using scroll wheel) or Slice Navigation (using the scroll wheel). YOu can toggle the structure on / off using the overlay controls below and also change the transparency of the contours. In the above image we show the two contours of mandible.